Bank Customer Churn Analysis Dashboard Report

Project Overview

The objective of this project was to analyze customer churn behavior in a banking dataset using Microsoft Power BI. The dashboard provides insights into customer demographics, geography, account activity, tenure, products, balance, salary, and credit score to help identify churn patterns and support customer retention strategies.

Data Preparation and Cleaning

Validated data quality, checked missing values, verified data consistency, standardized fields, created age groups, salary groups, balance groups, and credit score groups, and developed DAX measures for churn analysis.

Dashboard KPIs

Total Customers: 10,000
Churned Customers: 2,037
Churn Rate: 20%
Average Balance: 76K

Customer Churn Dashboard Analysis

1. Customer Churn by Gender: Female customers exhibit slightly higher churn than male customers.

2. Churn Distribution by Age Group: Customers aged 36–50 contribute the highest churn count.

3. Churn Rate by Number of Products: Customers holding 3 or more products show significantly higher churn rates.

4. Churned Customers by Geography: Germany records the highest churn compared to France and Spain.

5. Churn Rate by Customer Tenure: Churn generally decreases as customer tenure increases.

6. Churn Rate by Activity Status: Inactive customers are more likely to churn than active customers.

Financial Insights of Customer Churn

1. Churn Rate by Customer Balance: Higher balance groups show higher churn tendencies.

2. Churn Rate by Salary Group: Salary has limited influence on churn.

3. Churn Rate by Credit Score Group: Customers with lower credit scores exhibit slightly higher churn.

4. Churn Rate by Products and Activity Status: Customers with multiple products and inactive status show elevated churn risk.

5. Churn Rate by Age Group and Activity Status: Older inactive customers have the highest churn

rates.

Key Influencers Analysis

Power BI Key Influencers visual was used to identify major churn drivers. Key influencers include Number of Products, Activity Status, Geography, Age Group, and Credit Score. These factors have the strongest impact on customer churn behavior.

Interactive Features

The dashboard includes interactive slicers for Geography, Gender, and Activity Status, allowing users to dynamically explore customer segments and churn behavior.

Tools Used

Microsoft Power BI
Power Query
DAX (Data Analysis Expressions)
Data Modeling
KPI Cards
Maps and Interactive Visualizations
Key Influencers Visual

Business Recommendations

Focus retention campaigns on inactive customers, monitor high-value customers, improve engagement for customers with multiple products, and implement targeted strategies for high-risk geographic and demographic segments.

Conclusion

The Bank Customer Churn Analysis Dashboard successfully identifies churn patterns and key risk factors. The dashboard enables stakeholders to make data-driven decisions that support customer retention, improve engagement, and reduce churn.